Thermal Intertwiners

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Constructing semiclassical states of quantum geometry which resemble a smooth macroscopic Riemannian geometry on large enough scales continues to remain a challenge. Moreover in any theory of quantum gravity one should be able to provide a description of geometric states which correspond to systems at finite temperature. This would have implications, for instance, for the description of black hole horizons in quantum gravity. In loop quantum gravity, $U(N)$ coherent states (coherent intertwiners) provide one avenue towards constructing such states. In this work we show that the structure of these coherent intertwiners allows one to make contact with a thermal description of states of quantum geometry *without needing to introduce any extra ingredients into the picture*. We show how to define, via a Boguliubov transformation, thermal states for each face of such an intertwiner. We then argue that asking such faces to be glued to one other implies the presence of thermal equilibrium between different faces and an equipartition property for thermal intertwiners.

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I. INTRODUCTION

Quantum geometry must necessarily be an integral part of any theory of quantum gravity. By "quantum geometry" we refer to the structures which replace classical geometric concepts of lines, surfaces and volumes at the Planck scale. There is ample evidence to suggest that the spacetime which we perceive as a smooth manifold on nearly all scales, from subatomic to cosmological, must give way to a discrete, fluctuating geometric "foam" at the Planck scale. It is natural to expect that this quantum "foam" must have a quantum mechanical description and consistency requires that at sufficiently large scales the quantum mechanical description should begin to coincide with the classical picture of a smooth manifold. In

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quantum mechanics such states whose description is *as close as possible* to classical states are known as coherent states.

A second essential aspect of any picture of quantum geometry should be a concept of thermal states which correspond to fluctuating geometries. It is by now standard lore that gravity and thermodynamics are inextricably intertwined. Half a century ago, Bekenstein realized that black holes should be thermal objects which possess an entropy and a temperature. It was realized soon after by Hawking, Carter, Bardeen and others a black hole should radiate at a temperature proportional to the surface gravity of its horizon and that black holes should obey a set of laws of black hole thermodynamics in complete analogy with ordinary thermal systems. One might even go so far as to argue, following Jacobson , Padmanabhan and Verlinde , that Einstein's field equations are a consequence of imposing local thermal equilibrium between the degrees of freedom of geometry and matter. These developments suggest that in any theory of quantum gravity it should be possible to construct thermal states of quantum geometry.

Combining the above two insights implies that any theory of quantum gravity must permit the construction of *thermal coherent states* of quantum geometry.

One framework which permits the construction of such a quantum geometry is loop quantum gravity (LQG). In LQG there exist several candidate constructions of coherent states of geometry which are *peaked* around classical geometries with given values of geometric parameters such as the amount of intrinsic and extrinsic curvature of a surface. The construction of thermal states in LQG, however, has not been as successful as the coherent state program. In particular it would be desirable to be able to construct thermal coherent states of spin networks which correspond to semiclassical geometries at non-zero temperature. Some progress has been made towards this goal in the Group Field Theory (GFT) program , which can be viewed a second quantization of LQG.

The layout of this work is as follows. In section IV we review the construction of the classical phase space the and quantum state space for a single N-valent vertex, which could be part of a larger spin network, and recall how the degrees of freedom living on the faces of the polyhedron can be described using two independent harmonic oscillators. We then describe the construction of the spinorial version of the classical LQG phase space. This will be useful in the construction of coherent states in the following sections. In section II we explain how one can construct thermal states and coherent states for a single harmonic oscillator and how the two notions can be combined to give us thermal coherent states - i.e. states which obey the coherence property required for them to approximate classical behavior and also possess thermal properties parameterized by a temperature. In section IV C we begin by reviewing the construction of $U(N)$ coherent intertwiners which correspond to semiclassical states of polyhedral geometries. We then point out that the Schwinger boson representation which is utilized in constructing these states has all the necessary ingredients required to construct thermal states for each face of a polyhedron by performing a simple Boguliubov transformation on the face variables. The Boguliubov transformation depends on a parameter θ which can be identified with the temperature of the state. *A priori* there is no reason to take θ to be the same of all faces of the polyhedron. However, we argue that if this is not the case it becomes problematic to identify the resulting geometry with a semiclassical polyhedron which satisfies the closure condition on face normals. If we generalize from polyhedra to arbitrary two dimensional surfaces constructed by gluing quanta of area together then this line of reasoning suggests that such states represent semiclassical states of geometry only when the temperature parameter is the same for each area quantum, implying the requirement of a thermal equipartition property on semiclassical states of quantum geometry. We conclude with discussion of these results and some speculations.

II. THERMAL COHERENT STATES

A. Coherent states

Coherent states are the semi-classical states of quantum theory. The coherent state minimizes the uncertainty relation by minimizing and equating both Δx and Δp. The coherent states $|\alpha\rangle$ of a harmonic oscillator are eigenvectors of the annihilation operator:

$$a|\alpha\rangle = \alpha|\alpha\rangle \tag{1}$$

The state $|\alpha\rangle$ can also be created by acting with the displacement operator $D(\alpha)$ on the ground state:

$$|\alpha\rangle = D(\alpha)|0\rangle \tag{2}$$

where the displacement operator $D(\alpha)$ is given by

$$D(\alpha) = \exp(\alpha a^\dagger - \alpha^* a) \tag{3}$$

The coherent state $|\alpha\rangle$ can be expressed as a superposition of the energy eigenstates and it is given as :

$$|\alpha\rangle = e^{-\frac{|\alpha|^2}{2}} \sum_{n=0}^{n=\infty} \frac{\alpha^n}{\sqrt{n!}}|n\rangle \tag{4}$$

What makes these states so special is the fact that the real and imaginary components of the parameter α can be interpreted as the position and momentum of a *classical* harmonic oscillator as shown in Figure 1. Moreover under time evolution coherent states remain coherent, *i.e.* the property of classicality of a coherent state is maintained over time. It would not be sufficient to have states which were "coherent" (saturated the uncertainty bound) at a single time if time-evolution would cause that coherence to dissipate.

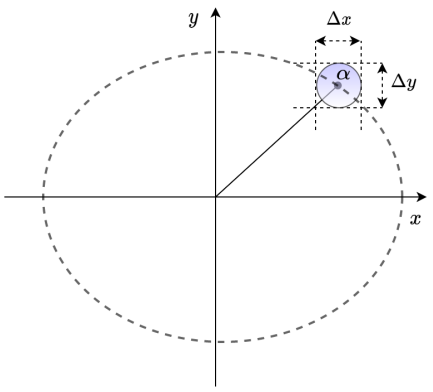


FIG. 1. The parameter α which labels the coherent state $|\alpha\rangle$ encodes the phase space co-ordinates of an ordinary classical oscillator. Time-evolution of the coherent state under the Schrödinger Hamiltonian yields a trajectory for α which is precisely that of the classical oscillator.

In addition to the property of coherence, as described above, we would like to endow our states with a thermodynamic description. It is well known how to construct "thermal states" in many body quantum mechanics and in quantum field theory. Is it possible to construct thermal states of quantum geometry which are also coherent? This would correspond to quantum states which describe the behavior of semi-classical geometries at finite temperature. Examples of such systems are the apparent horizon seen by an observer in a Rindler spacetime, or the "real" horizon seen by a static observer located outside a Schwarzschild black hole. To make progress towards this stage we first discuss the construction of thermal states and then thermal coherent states in the following two subsections.

B. Thermofield Double States

Traditionally there are two principle methods [1] which are used to construct thermal ensembles in quantum theory. The first method rests upon the notion of a density matrix ρ which describes the state of a system at a given time. The condition of thermal equilibrium is imposed quantitatively by the following two constraints on ρ. These are:

a) **Time independence:** The thermal state should be time-independent:

$$i\frac{\partial\rho(t)}{\partial t} = [\hat{H}, \rho(t)] = 0 \tag{5}$$

b) **Entropy maximization (Second law):** When the system is in thermal equilibrium, small variations in the state should not change the entropy of the system:

$$\delta S[\rho] = 0 \tag{6}$$

With these two assumptions in hand one can show [1] that at thermal equilibrium the density matrix of a system, in the canonical ensemble, is by the following expression:

$$\rho(\beta) = \frac{1}{Z}\exp\left(-\beta\hat{H}\right), \tag{7}$$

where $Z = \mathrm{Tr}[\exp(-\beta\hat{H})]$ is a normalization constant which we also recognize to be the partition function of the system. Here $\hat{H}$ is the Hamiltonian of the system and $\beta = (k_B T)^{-1}$ is the inverse temperature in Boltzmann units. Given any observable $\hat{A}$ the corresponding thermal expectation value is given by:

$$\langle\hat{A}\rangle = \mathrm{Tr}[\rho\hat{A}] \tag{8}$$

In analogy with the Gibbsian description of classical thermal equilibrium one can write down an expression for the entropy of a system described by a density matrix ρ:

$$S_{vN} = -k_B\mathrm{Tr}[\rho\ln\rho] \tag{9}$$

also known as the von-Neumann entropy.

The second approach, known as thermofield dynamics, starts by asking the question about whether or not it is possible to construct a state $|0(\beta)\rangle$ which represents the thermal ground state of a given system, such that thermal expectation values of any observables can be obtained by taking the expectation of the associated operator in the thermal ground state:

$$\langle\hat{A}\rangle = \langle 0(\beta)|\hat{A}|0(\beta)\rangle \tag{10}$$

Now, morally speaking, it might appear that is no real difference between (8) and (10) since for a pure state $\rho = |\psi\rangle\langle\psi|$ and the expression (8) reduces to (10) using the cyclic property of the trace:

$$\mathrm{Tr}[\rho\hat{A}] = \mathrm{Tr}[|\psi\rangle\langle\psi|\hat{A}] = \langle\psi|\hat{A}|\psi\rangle$$

However, there is a crucial difference due to the fact that the density matrix in (8) need not be that of a pure state, whereas in (10) we are requiring that the state $|0(\beta)\rangle$ must be a pure state. It turns out the second construction is possible if one introduces a *second* auxiliary Hilbert space which is a copy of the original system and the thermal state $|0(\beta)\rangle$ then lives in the tensor product space. As shown by Takahashi and Umezawa [2] we can define an extended Hilbert space:

$$\mathcal{H} = \mathcal{H}_L \otimes \mathcal{H}_R \tag{11}$$

where $\mathcal{H}_L$ and $\mathcal{H}_R$ stand for the ordinary Hilbert spaces for the zero temperature theory on the left hand side and the right hand side, respectively. We define a_L and $a_L^\dagger$ as the annihilation and creation operators acting on $\mathcal{H}_L$ and likewise for $\mathcal{H}_R$. These operators satisfy the commutation relation:

$$[a_L, a_L^\dagger] = [a_R, a_R^\dagger] = 1 \tag{12}$$

$$[a_L, a_R] = 0 \tag{13}$$

Given the form of (10) it turns out to be fairly straightforward to show (see section B) that the state $|0(\beta)\rangle$ can be written in terms of the energy eigenbasis of $\mathcal{H}_L$ and $\mathcal{H}_R$ as follows:

$$|0(\beta)\rangle = \frac{1}{\sqrt{Z(\beta)}} \sum_n e^{-\beta E_n/2} |n, \tilde{n}\rangle, \tag{14}$$

where $|n\rangle, |\tilde{n}\rangle$ are eigenstates of the number operators $a_L^\dagger a_L$ and $a_R^\dagger a_R$ and in the expression above $\tilde{n} = n$. The state $|0(\beta)\rangle$ is also known as the "thermo-field double" (TFD) state. $Z(\beta)$ is the partition function for the thermal state defined on a single copy of $\mathcal{H}_0$ (7).

As explained in section C the state $|0(\beta)\rangle$ can be obtained from the doubled oscillator vacuum $|0, \tilde{0}\rangle$ by the action of a unitary operator:

$$|0(\beta)\rangle = U(\beta)|0\rangle_L|0\rangle_R \tag{15}$$

The operator $U(\beta)$ is given by:

$$U(\beta) = \exp[\theta(a_L^\dagger a_R^\dagger - a_L a_R)] \tag{16}$$

where θ is related to β by:

$$\cosh(\theta) = (1 - e^{-\beta w})^{-\frac{1}{2}} \tag{17}$$

$$\sinh(\theta) = (e^{\beta w} - 1)^{-\frac{1}{2}} \tag{18}$$

This transformation of states also induces a transformation of the creation and annihilation operators. The operator transformation is given by the Boguliubov transformation:

$$a_L(\beta) = U(\beta) a_L U^\dagger(\beta) = \cosh(\theta) a_L - \sinh(\theta) a_R^\dagger \tag{19}$$

$$a_R(\beta) = U(\beta) a_R U^\dagger(\beta) = \cosh(\theta) a_R - \sinh(\theta) a_L^\dagger \tag{20}$$

With these "thermal" ladder operators on hand we can now construct thermal number operator states:

$$|n_1, n_2\rangle_\beta = \frac{(a_L^\dagger(\beta))^{n_1} (a_R^\dagger(\beta))^{n_2}}{\sqrt{n_1!}\sqrt{n_2!}} |0(\beta)\rangle \tag{21}$$

The state (15) has an important and interesting property which can be seen as follows. The exponential in (16) when acting on the vacuum $|0\rangle_L|0\rangle_R$ can be expressed as [3]:

$$\begin{aligned} U(\beta)|0\rangle_L|0\rangle_R &= \exp\left[\theta(a_L^\dagger a_R^\dagger - a_L a_R)\right] |0\rangle_L|0\rangle_R \\ &= \frac{1}{\cosh(\theta)} \exp\left[\tanh(\theta) a_L^\dagger a_R^\dagger\right] |0\rangle_L|0\rangle_R \\ &= \frac{1}{\sqrt{1 - e^{-\beta\hbar\omega}}} \sum_{n=0}^{\infty} e^{-n\beta\hbar\omega/2} |n\rangle_L|n\rangle_R \end{aligned} \tag{22}$$

In the limit of infinite temperature $\beta \to 0$, the state $|0(\beta)\rangle$ reduces to $\sum_n |n, n\rangle$ which is (upto a normalization factor) the maximally entangled state in the doubled Hilbert space $\mathcal{H}$. It is a remarkable fact that mixed thermal states in one Hilbert space can be mapped to highly entangled pure state in a doubled Hilbert space and this hints perhaps to the existence of a deeper connection between thermodynamics and entanglement.

C. Thermal Coherent States

We now have in hand the two ingredients we need to construct thermal coherent states. On the one hand we have the displacement operator (3) which tells us how to construct a coherent state $|\alpha\rangle$ labeled by a parameter α *given* a ground state $|0\rangle$. On the other hand we have the "thermalizing" operator (16) which takes the ground state of a doubled Hilbert space $|0, 0\rangle$ and converts it into a thermal ground state $|0(\beta)\rangle$ as shown in (15). The way to create thermal coherent states is now seemingly clear. Create the thermal vacuum using the operator (16) and *then* act on the thermal vacuum with the displacement operator (3) to obtain a thermal coherent state.

However, there is one minor issue. The coherent state construction requires only the original Hilbert space, whereas the thermal state construction requires a doubled Hilbert space. The solution to this problem is straightforward. We simply double the Hilbert space *before* applying the coherent state transform. The thermal coherent state is then given by [3] :

$$|TC\rangle = U(\beta)|\alpha\rangle_L|\gamma\rangle_R = U(\beta) D(\alpha, \gamma)|0\rangle_L|0\rangle_R, \tag{23}$$

where the operator $D(\alpha, \gamma)$ is now a displacement operator acting on the doubled Hilbert space and is simply given by the product of the the displacement operators on two copies of the original Hilbert space:

$$D(\alpha, \gamma) = D_L(\alpha) D_R(\gamma). \tag{24}$$

Note that since we have two *independent* coherent states we need two sets of the classical parameters α and γ.

There is a second route one can follow. Namely rather than constructing thermal coherent states $|TC\rangle$, we construct coherent thermal states $|CT\rangle$, *i.e.* we first construct the thermal ladder operators $a^\dagger(\beta)$ and $a(\beta)$ (19) using the operator $U(\beta)$ and then construct coherent states using the displacement operator built with these thermalized ladder operators. These two approaches can be summarized as: displace then thermalize (for $|TC\rangle$) or thermalize then displace (for $|CT\rangle$). In what follows we will be using the latter approach to construct coherent thermal states for $\mathfrak{su}(2)$ using which one can immediately construct coherent thermal intertwiners.

III. $\mathfrak{su}(2)$ THERMAL COHERENT STATES

In this section we will show how one can construct thermal coherent states for a particle with a given spin j. For this first we will review the Schwinger boson representation of the $\mathfrak{su}(2)$ algebra, following which we can use the procedure outlined in the previous section to construct thermal coherent states.

A. Schwinger Boson Construction

The Schwinger boson representation of the Hilbert space of a particle with spin j plays a central role in this work. We exploit the fact that one can express the Hilbert space of a *single* spin j particle in terms of *two* simple harmonic oscillators as follows. The $\mathfrak{su}(2)$ algebra is spanned by the operators J_x, J_y, J_z. We introduce two sets of ladder operators $a, a^\dagger, b, b^\dagger$ in terms of which the $\mathfrak{su}(2)$ operators can be expressed as follows:

$$J_z = \frac{1}{2}(a^\dagger a - b^\dagger b); \quad J_+ = a^\dagger b; \quad J_- = b^\dagger a \tag{25}$$

where $J_\pm = J_x \pm i J_y$ are the usual angular momentum raising and lowering operators and $a, a^\dagger, b, b^\dagger$ satisfy the familiar harmonic oscillator commutation relations:

$$[a, a^\dagger] = [b, b^\dagger] = 1; \quad [a, b] = [a, b^\dagger] = 0 \tag{26}$$

It is easy to check that the operators (25) defined in this way satisfy the usual $\mathfrak{su}(2)$ algebra: $[J_i, J_j] = i\epsilon_{ijk} J_k$.

States in the Schwinger boson representation are given by acting on the harmonic oscillator Fock vacuum with the number operators for the two oscillators:

$$|n_a, n_b\rangle = \frac{(a^\dagger)^{n_a}(b^\dagger)^{n_b}}{\sqrt{n_a! n_b!}}|0,0\rangle \tag{27}$$

The operator for the total energy of such a state is given by:

$$\mathcal{E} = \frac{1}{2}(a^\dagger a + b^\dagger b), \tag{28}$$

with eigenvalues:

$$\mathcal{E}|n_a, n_b\rangle = \frac{1}{2}(n_a + n_b)|n_a, n_b\rangle \tag{29}$$

Similarly the eigenvalues of the J_z operator defined in (25) are given by:

$$J_z|n_a, n_b\rangle = \frac{1}{2}(n_a - n_b)|n_a, n_b\rangle \tag{30}$$

This tells that we can identify $n_a - n_b$ as the magnetic quantum number m for an angular momentum state $|j; m\rangle$. Now by explicit calculation using (25) and the definition of the energy operator (28) one can verify that:

$$J^2 = \mathcal{E}(\mathcal{E} + 1), \tag{31}$$

where $J^2 = J_x^2 + J_y^2 + J_z^2$ is the $\mathfrak{su}(2)$ Casimir with eigenvalue $j(j+1)$ when acting on the state $|j; m\rangle$. This tells us that we can identify the eigenvalue of the energy operator $\mathcal{E}$ with the angular momentum quantum number j. So finally we see that the states of the harmonic oscillator space are in one-to-one correspondence with the usual angular momentum eigenbasis:

$$|j; m\rangle \equiv |n_a, n_b\rangle; \quad j = \frac{1}{2}(n_a + n_b); \; m = \frac{1}{2}(n_a - n_b). \tag{32}$$

j	m	n_a	n_b	$\lvert n_a, n_b\rangle$
1/2	+1/2	1	0	$\lvert 1,0\rangle$
	-1/2	0	1	$\lvert 0,1\rangle$
1	+1	2	0	$\lvert 2,0\rangle$
	0	1	1	$\lvert 1,1\rangle$
	-1	0	2	$\lvert 1,0\rangle$
3/2	+3/2	3	0	$\lvert 3,0\rangle$
	+1/2	2	1	$\lvert 2,1\rangle$
	-1/2	1	2	$\lvert 1,2\rangle$
	-3/2	0	3	$\lvert 0,3\rangle$

TABLE I. Mapping between angular momentum states in the usual representation in terms of $|j; m\rangle$ and states in the Schwinger boson representation $|n_a, n_b\rangle$

In the next section we will see that j is a measure of the total area associated with an edge of a spin network, with the precise expression depending on which of the two prescriptions in (38) is used for the area operator.

The physical picture of the Schwinger boson states is the following. A single particle of spin j is described by a pair of free harmonic oscillators with the constraint that the *total* energy $\mathcal{E}$ of both oscillators is fixed and is related to given by j as seen in expressions (32) and (29). The highest weight state $|j; j\rangle$ with $m = j$ corresponds to having one oscillator (say A) at the highest energy ($\mathcal{E} = j$) and the other (say B) in the ground state. As we act on the spin with the spin lowering operator J_- we lower the energy of the A oscillator by one unit and increase the energy of the B oscillator by one unit. When we reach the lowest weight state $|j; -j\rangle$ with $m = -j$ the A oscillator is in the ground state while the B oscillator is in the highest energy $\mathcal{E}$ state. This construction can be viewed as mapping a spin j system to a collection of $2j$ spin 1/2 particles with n_a spin up particles and n_b spin down particles as shown in Figure 2.

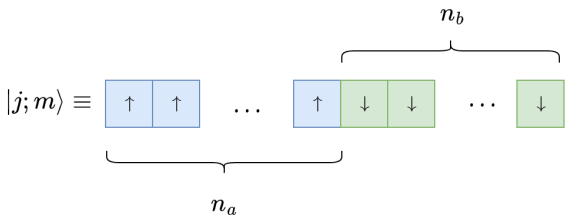


FIG. 2. Angular momentum state $|j; m\rangle$ can be mapped to a system of $2j$ spin 1/2 particles.

B. $\mathfrak{su}(2)$ Thermal Coherent State

Given the mapping between a pair of bosonic oscillators and the $\mathfrak{su}(2)$ algebra we can now put to use the techniques of the previous sections to construct thermal coherent states for a particle with spin j. First let us re-

mind ourselves how to construct coherent states for spin j. Recall that the coherent state for a single oscillator is of the form given by (4). We can easily generalize this to build a coherent state for a *pair* of oscillators. Following [4, Appendix B] we can write:

$$|z_a, z_b\rangle = e^{-(|z_a|^2+|z_b|^2)/2} \sum_{n_a,n_b} \frac{z_a^{n_a} z_b^{n_b}}{\sqrt{n_a! n_b!}} |n_a, n_b\rangle, \quad (33)$$

which is simply the tensor product of two copies of (4) with parameters z_a and z_b respectively labeling the two coherent states and the state $|n_a, n_b\rangle$ is given by (27). To obtain coherent states for a fixed spin j we simply need to project out those components which do not satisfy the constraint $2j = n_a + n_b$. If we write the projector as P_j, then after a little bit of algebra (see section D for the details) one finds that:

$$P_j|z_a, z_b\rangle = e^{-(|z_a|^2+|z_b|^2)/2} z_a^{2j} |j; z\rangle, \quad (34)$$

where $|j; z\rangle$ is given by:

$$|j; z\rangle = \sum_{m=-j}^{j} \frac{z^{j-m}}{\sqrt{(j+m)!(j-m)!}} |j; m\rangle, \quad (35)$$

which is (upto normalization) the $\mathfrak{su}(2)$ coherent state. Using (27) we can rewrite the above as:

$$|j; z\rangle = \sum_{m=-j}^{j} \frac{z^{j-m} (a^\dagger)^{j+m} (b^\dagger)^{j-m}}{(j+m)!(j-m)!} |0, 0\rangle \quad (36)$$

Using this we can write down the expression for $\mathfrak{su}(2)$ thermal coherent states:

$$|z(\beta)\rangle = \sum_{m=-j}^{j} \frac{z^{j-m} (a^\dagger(\beta))^{j+m} (b^\dagger(\beta))^{j-m}}{(j+m)!(j-m)!} |0(\beta)\rangle, \quad (37)$$

where $a^\dagger(\beta), b^\dagger(\beta)$ are the thermalized ladder operators (19) and $|0(\beta)\rangle$ is the thermofield double ground state (14).

IV. $U(N)$ COHERENT STATES IN LQG

LQG is based on the premise that one must take the background independence of Einstein's general theory of relativity seriously in any attempt to construct a theory of quantum gravity. LQG is built upon the seminal realization by Abhay Ashtekar that it is possible to recast General Relativity as a theory of a non-Abelian $SL(2, \mathbb{C})$ connection A^I_μ, and a $\mathfrak{sl}(2, \mathbb{C})$ valued one form E^J_μ. One can then proceed with the usual $3 + 1$ decomposition of the resulting Einstein-Hilbert-Palatini action and obtain a phase space whose points are labeled by a triad e^i_a (here $a, b \in \{1, 2, 3\}$ is a spatial index and $i, j \in \{1, 2, 3\}$ are $\mathfrak{su}(2)$ indices) and a $SU(2)$ connection ω^j_b. The final Hamiltonian takes the form of a constraint and is accompanied by two other constraints. These are the Gauss constraint which generates $SU(2)$ rotations of the triad field e^i_a and the diffeomorphism constraint which ensures that diffeomorphism symmetry of the spatial 3-manifold is respected.

Gauge invariant observables can be defined in terms of holonomies of the $SU(2)$ connection around closed loops and fluxes of the triad field across 2 dimensional surfaces. In terms of these observables one can construct an overcomplete set of basis states which solutions of the Gauss and diffeomorphism constraints. Elements of this basis are spin-networks, arbitrary graphs with edges labeled by representations of $SU(2)$ and vertices labeled by tensors known as "intertwiners" which serve to "tie up" all the edges coming into any given vertex. A spin network state is specified in terms of the triple: $\{\Gamma, \{j_i\}, \{I_{v_k}\}\}$, where Γ is a directed graph consisting of N_e edges and N_v vertices; $\{j_i; i \in 1 \dots N_e\}$ are $SU(2)$ group representation labels assigned to each edge and $\{I_{v_k}; k \in 1 \dots N_v\}$ are intertwining operators assigned to each vertex.

Classical geometrical observables such as lengths, areas and volumes are replaced by operators which act on the state space consisting of spin networks. Area operators act on the edge labels and volume operators act on the intertwiners. With each edge one can associate a quantum of area which is the area of a surface transversally intersected by that edge. With each vertex one can associate a quantum of volume which corresponds to the volume of a polyhedron centered on that vertex and which has as many faces as the degree[1] of that vertex.

A. Area and Volume Operators

The spin labels on each edge carry physical information about the geometry represented by the spin network state. Consider an abstract surface pierced by a single edge carrying a spin j. Then from LQG we know that that edge endows the surface with a quantum of area[2] which depends on the spin label. There are two proposals for the area associated with a puncture. These are:

$$A_{RS} = l_p^2 \sqrt{j(j+1)} \quad (38a)$$

$$A_{FL} = l_p^2 j \quad (38b)$$

The first is the proposal of Rovelli and Smolin that the area of a puncture is proportional to the Casimir of the spin j. The second is the proposal of Freidel and Livine (and the one we will use in this work) where the area

[1] In graph theory, the "degree" of a given vertex is the number of edges attached to that vertex

[2] It rather easy in the LQG framework to see how this comes about. The interested reader is referred to Sec 6.1 of [5] for a gentle treatment.

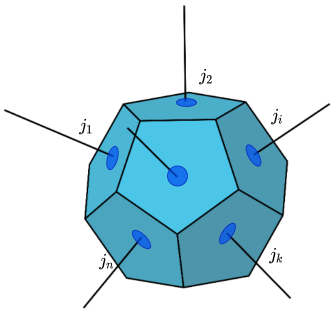


FIG. 3. An example of a spin-network vertex, with the associated polyhedron. The faces of the polyhedron are transverse to the edges attached to the vertex. The vertex itself lies in the interior of the polyhedron. Each edge punctures one of the faces and assigns it a quantum of area depending on the spin rep labelling the edge.

is simply proportional to the highest eigenvalue of the spin j representation. In the Freidel-Livine picture we can view any vertex of a spin network as being dual to a polyhedron[3] whose faces are pierced by edges adjacent to that vertex as in Figure 3. In order that the faces associated with each edge, adjacent to a given vertex v, can be glued together consistently to form a polyhedron without any "gaps" the following constraint must be satisfied by the spin-network state at each vertex:

$$(\vec{J}_1 + \vec{J}_2 + \dots \vec{J}_n)|\Psi_\Gamma\rangle = 0, \tag{39}$$

where $\vec{J}_1, \dots, \vec{J}_n$ are the angular momentum operators associated with each edge adjacent to the given vertex. This is a statement of the Gauss constraint which says that the (physical) spin network state should be invariant under the action of $SU(2)$ gauge transformations acting on its vertices. Now the Hilbert space of all the edges adjacent to a given vertex can be expressed as the tensor product:

$$\mathcal{H}_v = \mathcal{H}_1 \otimes \mathcal{H}_2 \otimes \dots \otimes \mathcal{H}_n, \tag{40}$$

where n is the number of edges attached to the vertex v. Imposing the constraint (39) now requires that we assign an invariant tensor to each vertex, *i.e.* an element of the tensor product space $\mathcal{H}_v$ which is invariant under $SU(2)$ rotations generated by (39). On this space of invariant tensors $\text{Inv}_{SU(2)}(\mathcal{H}_v)$ one can define a volume operator whose eigenvalues are the possible volumes of the allowed polyhedra dual to that vertex.

[3] It is probably not too difficult to construct a polyhedron whose face areas are given by A_{RS}. However, the presence of the square root makes it difficult to see how one would express the closure constraint for the face normals in terms of face areas with an expression as simple as Equation 39.

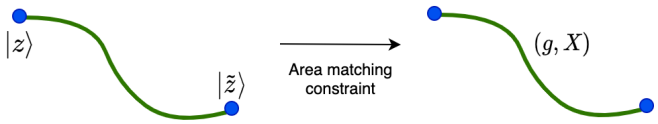


FIG. 4. In spinorial LQG we label each end of an edge with spinors $(|z\rangle, |\tilde{z}\rangle) \in \mathbb{C}^2 \times \mathbb{C}^2$. Applying the area matching constraint $\|z\|^2 - \|\tilde{z}\|^2 = 0$, we obtain the phase space variables (g, X), with $g \in SU(2)$ and $X \in \mathfrak{su}(2)$.

B. Spinorial LQG

The spinorial formulation of LQG was developed in a series of seminal papers by Freidel and Speziale [6, 7]. The classical phase space of a single spin network edge is given by $T^*[SU(2)] \simeq SU(2) \times \mathfrak{su}(2)$ - the cotangent bundle on $SU(2)$. Configuration variables are group elements $g \in SU(2)$ and the momenta are elements of the lie algebra $\mathfrak{su}(2)$. Recall that the Hilbert space of a *single* edge is given by (E4) the space of square integrable functions $\mathcal{L}^2[SU(2)]$ on $SU(2)$. This space can be viewed as the quantization of the classical phase space $T^*[SU(2)]$.

What was shown in the works by Freidel and Speziale was that one can obtain $T^*[SU(2)]$ by starting from the space $\mathbb{C}^2 \times \mathbb{C}^2$ whose elements are pairs of spinors $|z\rangle, |\tilde{z}\rangle$. From each spinor we can construct a three-vector as follows:

$$\vec{X} = \langle z|\vec{\sigma}|z\rangle; \quad \vec{\tilde{X}} = \langle \tilde{z}|\vec{\sigma}|\tilde{z}\rangle \tag{41}$$

where $\vec{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ are the three Pauli matrices. For a brief introduction to spinorial notation we refer the reader to Appendix F. These vectors are then to be thought as corresponding to the normals to the faces which are dual to a given edge (Figure 5). In the classical picture, the area of a face is given by the length of the vector normal to it. Thus, the vectors $\vec{X}, \vec{\tilde{X}}$ are the classical counterparts of the angular momentum operators $\vec{J}$ assigned to each end of an edge. These vectors satisfy the identity:

$$\|X\| = \langle z|z\rangle; \quad \|\tilde{X}\| = \langle \tilde{z}|\tilde{z}\rangle \tag{42}$$

We can construct the momentum variable - the element of $\mathfrak{su}(2)$ - associated with each end of a vertex by taking the corresponding face normals and contracting them with the Pauli matrices:

$$X = \vec{X} \cdot \vec{(\sigma)}; \quad \tilde{X} = \vec{\tilde{X}} \cdot \vec{\sigma} \tag{43}$$

We can construct the group element $g \in SU(2)$ corresponding to the holonomy along the given edge (and plays the role of the configuration variable of the classical phase space $T^*[SU(2)]$) as follows:

$$g(z, \tilde{z}) = \frac{|z\rangle[\tilde{z}| - |z]\langle\tilde{z}|}{\|z\|\|\tilde{z}\|} \tag{44}$$

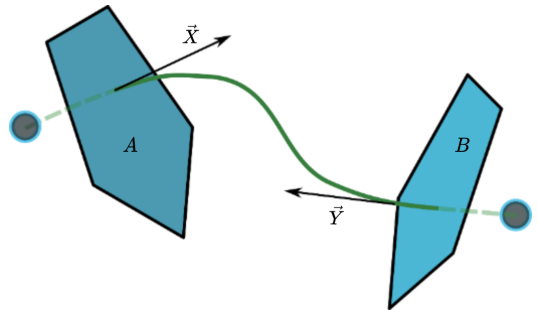


FIG. 5. The normal vectors to the faces at each end of an edge in terms of the spinors attached to each end.

Now, in order for the spin network to correspond to a triangulation of our background manifold, we must be able to glue the faces labelled A and B in Figure 5 to each other. This implies the area of the two faces must match. Since the area of a face is proportional to the length of the face normal, this implies the following *area matching constraint* on the two spinors $|z\rangle$ and $|\tilde{z}\rangle$:

$$\mathcal{M} = \langle z|z\rangle - \langle \tilde{z}|\tilde{z}\rangle = 0, \tag{45}$$

must hold. It is the imposition of this constraint which allows us to reduce the spinorial phase space $\mathbb{C}^2 \times \mathbb{C}^2$ to the phase space of a single edge $T^*[SU(2)]$.

In [6, 7] the authors showed the following correspondence:

$$T^*[SU(2)] \simeq (\mathbb{C}^2 \times \mathbb{C}^2)//U(1), \tag{46}$$

where the "double quotient" is understood as taking those elements of $\mathbb{C}^2 \times \mathbb{C}^2$ which live on the constraint surface determined by (45) and which Poisson commute with this constraint. It is understood that degenerate points, *i.e.* those for which $\|X\| = 0$, $\langle z|z\rangle = 0$, $\langle \tilde{z}|\tilde{z}\rangle = 0$ are excluded from both sides of the above relation.

We can now address what we set out to do - labeling the classical phase space of the intertwiner. For this purpose we simply need to label the ends of intertwiner legs which meet the given vertex with spinor $|z_i\rangle; i \in \{1, \ldots, n\}$ as shown in Figure 6.

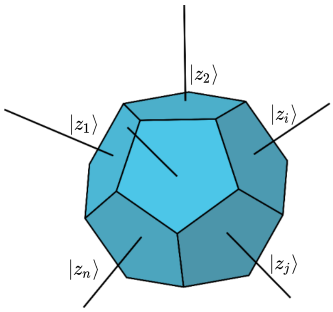


FIG. 6. The classical phase space of an intertwiner. Each leg is labelled by a spinor $|z_i\rangle \in \mathbb{C}^2$ where $i \in \{1, \ldots, n\}$.

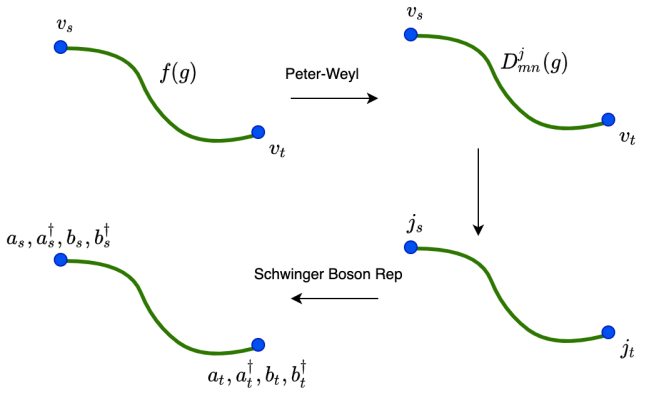


FIG. 7. Illustration of steps leading from cylindrical functions of edge holonomies to Schwinger Boson basis with pairs of ladder operators assigned to the source and target vertices respectively.

It is this set of spinors which provides us with the set of labels needed to parametrize a given $U(N)$ coherent state.

C. $U(n)$ Coherent Intertwiners

In order to construct coherent states for intertwiners we first use the Schwinger boson trick to write the spin associated with each face in terms of two oscillators. This step is illustrated in the lower half of Figure 7. There both of the spins at *both* ends of a given edge are mapped to their respective pair of oscillators. In our case we our only interested in a single quantum of spacetime volume which corresponds to a single n-valent single vertex, so we only care about what happens at *one* end of an edge which meets a given face of that intertwiner. The end result is illustrated in Figure 8.

We now define the following operators which act on *pairs* of faces:

$$E_{ij} = a_i^\dagger a_j + b_i^\dagger b_j. \tag{47}$$

These operators have a very simple physical interpretation. They destroy one quantum of type a and of one of type b living the jth face and create one quantum of type a and one of type b on the ith face. We can think of operator as a *hopping* term in the language of many body physics. It is clear that this operator is hermitian. What makes these operators useful is that they commute with the generator of global $SU(2)$ transformations generated by the *total* angular momentum operator $\vec{J} = \sum_{i=1}^n \vec{J}_i$:

$$\left[\vec{J}, E_{ij}\right] = 0 \quad \forall i, j. \tag{48}$$

Furthermore the set of operators E_{ij} with $i, j \in$

$\{1, 2, \ldots, n\}$ is closed under commutation [8]:

$$[E_{ij}, E_{kl}] = \delta_{jk} E_{il} - \delta_{il} E_{kj}, \tag{49}$$

which we recognize to be the Lie algebra of the group $U(n)$. If we choose to measure the area of a face with the Freidel-Livine prescription (38b) rather than the Rovelli-Smolin prescription (38a), then from (28) we see that for $i = j$, the operator $E_i \equiv E_{ii}$ corresponds to *twice* the area of the ith face. The operator $E = \sum_{i=1}^{n} E_i$ therefore corresponds to *twice* the total area of the polyhedron. The operators E_{ij} generate $U(n)$ transformations of the polyhedron which correspond to *area-preserving* diffeomorphisms. This property can be seen by noting that the action of E_{ij} on the jth face lowers j_j by $1/2$ and its action on the ith face raises the value of j_i by $1/2$, thus keeping the sum $\sum_{i=1}^{n} j_n$, which corresponds to the total area, a constant.

In order to define $U(n)$ coherent states we need one more ingredient. These are the $U(n)$ ladder operators given by:

$$F_{ij} = a_i b_j - a_j b_i; \quad F_{ij}^\dagger = a_i^\dagger b_j^\dagger - a_j^\dagger b_i^\dagger. \tag{50}$$

We define $|0)$ to be the Fock vacuum, which satisfies $F_{ij}|0) = 0$, $\forall i, j$. We can now define the $U(n)$ coherent states:

$$|J, z_i) = \frac{1}{\sqrt{J+1}} \frac{(F_{\mathbf{z}}^\dagger)^J}{J!} |0). \tag{51}$$

Here J measures the total area of the coherent polyhedron represented by this state; and $\mathbf{z} \equiv z_{ij}$ is a matrix defined in terms of $\{z_i\}$ which are the spinor labels assigned to the end of each edge adjacent to the given vertex as shown in Figure 6. As explained in the previous section, these spinors label a point in the classical phase space of n sided polyhedra. Using these spinors we define:

$$\mathbf{z} \equiv z_{ij} = [z_i | z_j\rangle \tag{52}$$

where the inner product in the above expression is defined in (F4). Using the definition of z_{ij} one can show that $z_{ij} = -z_{ji}$, i.e. that $\mathbf{z}$ is an anti-symmetric matrix. The quantity $F_{\mathbf{z}}^\dagger$ in (51) is defined as follows:

$$F_{\mathbf{z}} = \frac{1}{2} \sum_{ij} \bar{z}_{ji} F_{ij}; \quad F_{\mathbf{z}}^\dagger = \frac{1}{2} \sum_{ij} z_{ji} F_{ij}^\dagger. \tag{53}$$

In the definition of these coherent state (51) there is a parameter J. One can see that this parameter corresponds to the total area of the n sided polyhedron as follows. A straightforward computation shows that:

$$[E, F_{\mathbf{z}}^\dagger] = 2F_{\mathbf{z}}^\dagger \tag{54}$$

where $E = \sum_i E_i$; and $E_i := E_{ii}$ are the diagonal generators of the $\mathfrak{u}(n)$ algebra. Using this expression along with standard commutator identities one can show that:

$$[E, (F_{\mathbf{z}}^\dagger)^J] = 2J (F_{\mathbf{z}}^\dagger)^J; \quad \forall J \in \mathbb{N}. \tag{55}$$

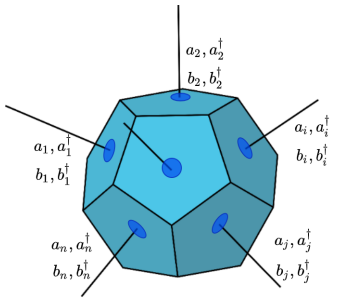


FIG. 8. Applying the Schwinger boson trick to each spin label on all the edges.

Applying this expression to the definition of the coherent states (51) one can show that:

$$E|J, z_i) = 2J|J, z_i) \tag{56}$$

From the discussion following (49) we know that eigenvalues of the operator E correspond to *twice* the area of the polyhedron. Therefore we can identify J in the above expressions as the total area of the polyhedron.

It can be shown explicitly [8] that these states (51) are indeed generalized coherent states in the sense of Perelomov [9] and are generated by the action of elements of $U(n)$ on the highest weight vector. As a result they are also minimum uncertainty states in that the following ratio: $\Delta E_{ij}/\langle E_{ij}\rangle \sim 1/\sqrt{J}$ vanishes in the limit of large values for the total area $A_{tot} = 2J$ of the polyhedron. Here:

$$\Delta^2 E_{ij} = \langle E_{ij} E_{ji}\rangle - \langle E_{ij}\rangle \langle E_{ji}\rangle \tag{57}$$

is the dispersion in the expectation value of the operators E_{ij}.

V. COHERENT THERMAL INTERTWINERS

We now have all the ingredients in hand in order to construct *thermal* $U(n)$ coherent intertwiners. One might think that the procedure to do would be the following:

a) construct the thermalizing operator $U(\beta)$ (16)

b) use $U(\beta)$ to obtain the thermalizing ladder operators as shown in (19)

c) use these thermalized ladder operators to construct the coherent thermal state.

However, there are some subtleties along the way we have to take care of. The first is the following. The construction of the $\mathfrak{su}(2)$ coherent thermal states (37) double state described in section II B requires a pair of oscillators. We already have these in the form of the oscillator pairs a_i, b_i assigned to each face of the polyhedron. However, now rather than just one spin we now have to define the thermal coherent state of n spins. This would suggest that

rather than a single inverse temperature β we now need n different values of the inverse temperature β_i, one for each face of the polyhedron. We could proceed by assuming that each face is at a different temperature and construct the global thermalizing operator:

$$\mathbf{U} = U(\beta_1) \otimes U(\beta_2) \otimes \ldots \otimes U(\beta_n), \tag{58}$$

where $U(\beta_i)$ is the thermalizing operator acting on the doubled Hilbert space of the ith puncture. What happens to the closure constraint under this transformation? Write G_X for the ordinary Gauss constraint on the single-copy intertwiner space and define its action on the doubled space by

$$\boldsymbol{G}_L = G_L \otimes I_R, \qquad \boldsymbol{G}_R = I_L \otimes G_R. \tag{59}$$

We begin with two conjugate coherent intertwiners and apply the thermal transformation:

$$\begin{aligned} |\Phi\rangle &= |J, z\rangle_L \otimes |\overline{J, z}\rangle_R, \\ |\Psi_\beta\rangle &= \mathbf{U}|\Phi\rangle. \end{aligned} \tag{60}$$

The original constraints annihilate $|\Phi\rangle$. Unitary conjugation therefore gives

$$\begin{aligned} \boldsymbol{G}_X|\Phi\rangle &= 0, \\ \boldsymbol{G}_X(\beta)|\Psi_\beta\rangle &= 0, \\ \boldsymbol{G}_X(\beta) &= \mathbf{U}\boldsymbol{G}_X\mathbf{U}^\dagger, \quad X = L, R. \end{aligned} \tag{61}$$

The thermal transformation carries closure into a constraint on the thermalized fluxes. To ask whether each heated copy also forms a closed geometry with respect to its original fluxes, we must examine $\boldsymbol{G}_X$ on $|\Psi_\beta\rangle$. These operators need not annihilate the heated state.

It is not sufficient to understand only how the individual face Hilbert space transform under the thermalizing operator because this is not a system of n *independent* spins. Rather the coherent state constructed via (51) relies on the operators F_{ij} which link pairs of different faces and do not transform in a simple manner under $U_{ij} = U(\beta_i) \otimes U(\beta_j)$.

We can circumvent these issues by making the assumption that the inverse temperature $\beta_i = \beta$ is the same for all faces. In this case the operators E_{ij} and F_{ij} *will* have a simple transformation under the global $\mathbf{U}$ and one can verify that the thermalized $E_{ij}(\beta)$ will continue to satisfy the $U(n)$ Lie algebra given in (49). The assumption of having the same temperature for all faces is not just a mathematical convenience. It has a deep physical significance. In the context of black hole thermodynamics, for instance, this condition would correspond to the assumption of equipartition of energy between the microstates which constitute the horizon macrostate for a given value of the area A.

By assuming same temperature for all the surfaces, the expression for $\mathbf{U}$ is given as

$$\mathbf{U} = exp\left[\sum_{i=1}^{N} \theta[a^\dagger_{i_L} a^\dagger_{i_R} - \; a_{i_L} a_{i_R} + b^\dagger_{i_L} b^\dagger_{i_R} - b_{i_L} b_{i_R}]\right] \tag{62}$$

For the common-temperature squeeze and the conjugate right-copy representation, the doubled state obeys the combined constraint

$$(G_L \otimes I_R - I_L \otimes \overline{G}_R)|\Psi_\beta\rangle = 0. \tag{63}$$

Here the bar denotes the conjugate representation. The residual flux of one copy is compensated by the other. This combined closure does not require either copy to be an ordinary singlet by itself.

The earlier one-sided construction, with an FL excitation only on the left, can be written as:

$$|J, z_i)_{\beta_L} = \frac{1}{\sqrt{J+1}} \frac{(\tilde{F}^\dagger_{\mathbf{z}_L})^J}{J!} |0(\beta)), \tag{64}$$

where $\tilde{F}^\dagger_{\mathbf{z}}$ is defined in terms of the thermalized $U(n)$ ladder operators:

$$\tilde{F}_{ij} = a_i(\beta) b_j(\beta) - a_j(\beta) b_i(\beta) \tag{65}$$

In this one-sided expression the right copy is vacuum with respect to the transformed right annihilation operators, not the original oscillators. The two-FL state in (60) additionally excites the conjugate right intertwiner; it is the preparation studied below.

The squeeze correlates geometric degrees of freedom in the two copies. Interpreting this entanglement as a wormhole would require additional geometric matching and gravitational dynamics; the doubled-state construction and combined closure alone do not establish such a geometry.

VI. TEMPERATURE AND CLOSED QUANTUM GEOMETRY

What happens to the geometry when we increase its temperature? The squeeze excites the face oscillators and spreads the state over different areas. We can now ask whether this growth survives when we require both copies to close under their ordinary fluxes.

We use the normalized four-face reference $|\phi\rangle = F^\dagger_{12}|0\rangle/\sqrt{2}$ in each copy, with conjugate pairing on the right. Its initial area is $J_{\text{in}} = 1$: faces 1 and 2 carry spin 1/2 and form a singlet, while faces 3 and 4 have zero area. The starting geometry is therefore closed and has zero volume. We measure temperature and final linear area by

$$T_* := \frac{k_B T}{\hbar\omega}, \qquad A = \frac{q}{2}, \tag{66}$$

where q is the final boson number in either copy. The initial area label remains fixed while the final area fluctuates.

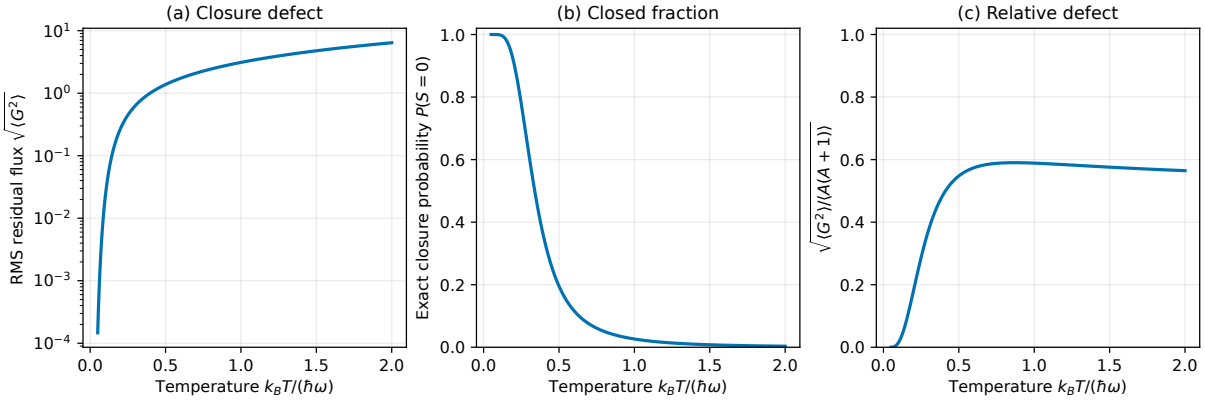


FIG. 9. Ordinary closure of either copy as temperature increases. The RMS residual grows and the singlet fraction falls. The relative residual in panel (c) is normalized by $\sqrt{\langle A(A+1)\rangle}$; its eventual decrease reflects growing area, while the absolute closure defect continues to rise.

A. Ordinary closure and separate singlet projection

A vanishing mean flux does not establish closure: nonzero resultant spins can have directions which average away. Their size is measured by

$$D_\beta = \langle \Psi_\beta | \boldsymbol{G}_L^2 | \Psi_\beta \rangle = \sum_{q,S} S(S+1) P_\beta(q,S), \tag{67}$$

with angular momenta in units of $\hbar$. Heating increases this ordinary closure defect, as shown in Figure 9. At $T_* = 1$, its RMS value is $\sqrt{D_\beta} = 3.10919$, and only 2.64065% of either copy lies in the singlet sector. At $T_* = 2$, these values are 6.41372 and 0.34037%. The transformed and combined constraints continue to hold exactly.

To impose ordinary closure on both copies, let $P_{X,0}$ project onto the single-copy singlet space and form

$$\begin{aligned} \mathcal{P}_0 &= P_{L,0} \otimes P_{R,0}, \\ p_\beta &= \langle \Psi_\beta | \mathcal{P}_0 | \Psi_\beta \rangle, \\ |\Psi_\beta^{\text{closed}}\rangle &= \frac{\mathcal{P}_0 |\Psi_\beta\rangle}{\sqrt{p_\beta}}. \end{aligned} \tag{68}$$

Both ordinary constraints now annihilate the state:

$$\boldsymbol{G}_L |\Psi_\beta^{\text{closed}}\rangle = \boldsymbol{G}_R |\Psi_\beta^{\text{closed}}\rangle = 0. \tag{69}$$

Closure fixes the resultant spin, but leaves the area free to fluctuate. Moreover the two copies are correlated: their coefficient matrices connect equal resultant-spin sectors. Selecting a singlet on the left selects a singlet on the right as well, so

$$p_\beta = \sum_q P_\beta(q, S=0). \tag{70}$$

The retained probability is the single-copy singlet fraction, rather than its square. It is positive at every finite temperature; the vacuum-vacuum component alone contributes $(1-e^{-b})^8 e^{-2b} > 0$, where $b = 1/T_*$. At zero temperature the original closed product returns.

B. Growth of area and volume

The closed state retains a strong response to temperature, as shown in Figure 10. Its conditional area distribution is

$$\mathcal{P}_\beta(A) = \frac{P_\beta(2A,0)}{p_\beta}. \tag{71}$$

At $T_* = 1$, its mean area is 2.78775 and its mean positive Rovelli–Smolin volume is 0.0723452. At $T_* = 2$, they rise to 6.47574 and 0.353506. Volume is evaluated by summing the four positive triple-grasp roots, in project units with $\gamma = 0.2375$ and $\hbar = 1$. Heating thus populates larger closed configurations and produces positive volume from an initially zero-volume reference.

Closure alone does not require four nonzero-area faces or nonzero volume. Figure 11 shows how these configurations become populated. At $T_* = 1$, the projected probabilities of these two properties are 18.61% and 14.08%; at $T_* = 2$, they reach 48.48% and 36.34%. The volume probability is obtained from the kernel of the positive volume operator.

The high-temperature area growth can be obtained analytically from the exact singlet-sector sums. With $R_\beta = \sum_A A P_\beta(2A,0)$, their leading terms are

$$p_\beta \sim \frac{7}{256} b^3, \qquad R_\beta \sim \frac{57}{512} b^2. \tag{72}$$

Their ratio gives

$$\langle A \rangle_{\text{closed}} \sim \frac{57}{14} T_*. \tag{73}$$

The closed fraction falls as T_*^{-3} while the area within that fraction grows linearly with temperature. Positive volume also increases over the calculated range, but its curve is nonlinear. On $1 \le T_* \le 2$, the area fit is $\langle A \rangle \simeq 3.70213 T_* - 0.96406$ with $R^2 = 0.999686$, distinct from the asymptotic slope 57/14. The volume's local

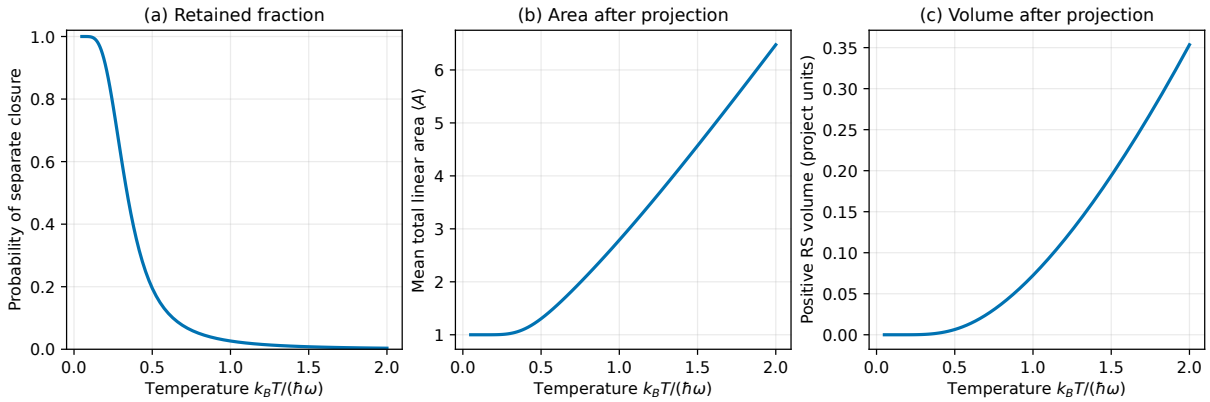


FIG. 10. Geometry after separate singlet projection. The retained fraction decreases with temperature, while the mean area and positive RS volume of the normalized closed state increase. Both ordinary Gauss constraints hold throughout. Area is in spin units and volume uses $\gamma = 0.2375$, $\hbar = 1$, in project units.

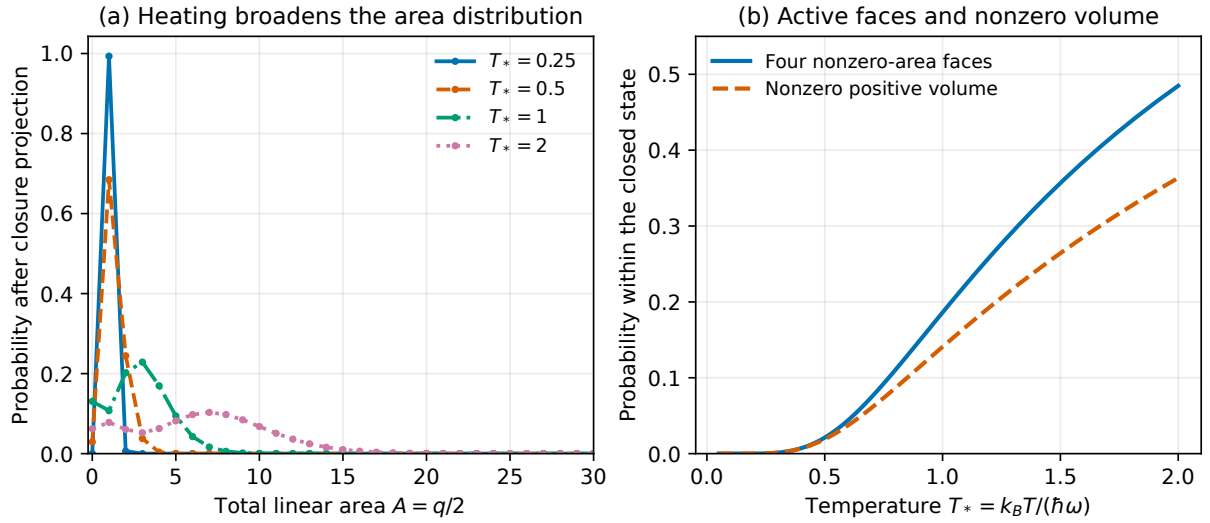


FIG. 11. The geometric content of the separately closed state. The area distribution broadens towards larger areas as temperature rises. The probabilities of four active faces and nonzero positive RS volume also increase; both are conditional on separate closure. Markers in panel (a) denote discrete area sectors.

slope rises from approximately 0.196 to 0.351 across that interval, so a constant linear high-temperature volume coefficient is not established. The sector-sum derivation is given in section A.

The closure calculation covers 125 temperatures and the projected geometry covers 85, over $0.05 \le T_* \le 2$. Occupation probabilities extend through $q = 160$ and volume traces through $A = 50$. Independent magnetic counting and spin recoupling agree; direct double projection verifies both Gauss constraints. Volume traces agree with the independently constructed closed-state catalogue through $A = 12$, and the largest bound on the omitted conditional mean-volume contribution is 5.4×10^{-12}. The area asymptote follows from exact polynomial sector sums rather than extrapolation of the numerical cutoff.

VII. THERMAL GEOMETRY AND EQUILIBRIUM

The closed state supplies temperature-dependent geometric distributions. What statistical mechanics can we extract from them? Measurements confined to the left copy are described by

$$\rho_L^{\text{closed}} = \mathrm{Tr}_R\, |\Psi_\beta^{\text{closed}}\rangle\langle\Psi_\beta^{\text{closed}}|. \tag{74}$$

This density matrix gives areas, volumes, correlations and their fluctuations, as well as the entropy of the reduced state. A derivative such as $d\langle A\rangle/dT$ measures an area response. To interpret it as a heat capacity, however, we must specify an energy operator and an equilibrium ensemble.

A canonical Gibbs family uses the same Hamiltonian

at every temperature:

$$\rho_\beta = \frac{e^{-\beta H}}{Z_\beta}, \qquad Z_\beta = \operatorname{Tr} e^{-\beta H}. \tag{75}$$

Only with this prescription do the standard relations

$$\mathcal{F} = -k_B T \log Z_\beta, \qquad C = \frac{d\langle H\rangle}{dT} = \frac{\langle H^2\rangle - \langle H\rangle^2}{k_B T^2} \tag{76}$$

apply. The relation $\tanh\theta = e^{-\beta\hbar\omega/2}$ fixes our preparation parameter; it does not by itself establish (75) for an excited, separately projected FL state.

A. An obstruction to a fixed Gibbs Hamiltonian

An energy law can be tested without calculating the complete entropy. If two eigenvectors of ρ_β are common to the family, their eigenvalues must satisfy

$$\log \frac{p_n(\beta)}{p_m(\beta)} = -\beta(E_n - E_m). \tag{77}$$

The inferred gap must be temperature independent. In particular, the ordering of two populations cannot reverse at positive temperature.

For the projected four-face reference, set $x = e^{-b}$ and $\eta = 1 - x$. In final integer-area sector K, the coefficient matrix is

$$\begin{aligned} C_K &= \eta^4 x^{K-1}[uX + vn_{12} + wI], \\ X &= F_{12}^\dagger F_{12}, \qquad n_{12} = n_1 + n_2, \\ u &= \eta^2/2, \qquad v = -\eta x/2, \qquad w = x^2. \end{aligned} \tag{78}$$

Here $n_i = a_i^\dagger a_i + b_i^\dagger b_i$. The commuting operators admit a common singlet recoupling basis. The reduced eigenvalues are the squared coefficient eigenvalues, divided by the same retained norm p_β.

The vacuum has $K = 0$, $X = n_{12} = 0$, while the input reference $|\phi\rangle = F_{12}^\dagger|0\rangle/\sqrt{2}$ has $K = 1$, $X = n_{12} = 2$. Their coefficients are

$$c_{\mathrm{vac}} = \eta^4 x, \qquad c_\phi = \eta^4(1 - 3x + 3x^2). \tag{79}$$

Consequently

$$\frac{p_{\mathrm{vac}}}{p_\phi} = \frac{x^2}{(1 - 3x + 3x^2)^2}. \tag{80}$$

At low temperature this ratio approaches e^{-2b}, suggesting that a fixed Gibbs Hamiltonian would place the vacuum $2\hbar\omega$ above the reference. Yet at $x = 1/2$, corresponding to $T_* = 1/\log 2$, the ratio is 4: the vacuum is more probable. No fixed gap reproduces both behaviors. The interference in the excited coefficients changes their relative preferences as temperature rises.

This excludes a temperature-independent single-copy canonical Hamiltonian for this projected reference family at the stated temperature. The argument concerns individual density-matrix eigenvalues, not area-sector probabilities which also include multiplicities. It does not exclude equilibrium in a larger interacting system. At any one temperature an effective operator can still be defined on the occupied support,

$$H_{\mathrm{eff}}(T) = -k_B T \log \rho_L^{\mathrm{closed}}(T) + c(T) I, \tag{81}$$

but its temperature dependence cannot be replaced by one fixed single-copy energy law.

B. Area conservation and geometric heating

Fixing area does not fix volume: changing face correlations can change the enclosed geometry. A number-conserving hopping Hamiltonian can therefore generate volume dynamics inside a fixed-area sector. An ensemble spanning several area sectors can also have a temperature-dependent mean area while its Hamiltonian conserves area, because the sector weights change with temperature. Actual dynamical exchange between individual-area sectors requires another system or a source.

The temperature used in section VI measures excitation relative to a fixed oscillator scale. It should not be identified directly with the Hawking temperature at infinity. For a Schwarzschild horizon,

$$T_H = \frac{\hbar c}{2k_B\sqrt{\pi A_H}}, \qquad A_H = \frac{\hbar^2 c^2}{4\pi k_B^2 T_H^2}, \tag{82}$$

so the horizon area decreases as its temperature increases. Our occupation-derived area instead grows with excitation. Relating these quantities requires a gravitational energy law and a relation between the oscillator scale and surface gravity.

VIII. SINGLET EXCHANGE BETWEEN CLOSED POLYHEDRA

How can a polyhedron exchange area while remaining closed? The FL algebra already supplies the required operation. For copy $X = L, R$, define

$$\begin{aligned} A_X &= \frac{1}{2}\sum_i (a_{iX}^\dagger a_{iX} + b_{iX}^\dagger b_{iX}), \\ F_{ij,X}^\dagger &= a_{iX}^\dagger b_{jX}^\dagger - a_{jX}^\dagger b_{iX}^\dagger. \end{aligned} \tag{83}$$

The singlet creator obeys

$$[A_X, F_{ij,X}^\dagger] = F_{ij,X}^\dagger, \qquad [G_X^a, F_{ij,X}^\dagger] = 0. \tag{84}$$

It raises area by one unit without violating closure. We can therefore transfer a singlet from one copy to the other, preserving both separate constraints.

A. Hamiltonian and area current

We examine the Hamiltonian

$$\begin{aligned} H &= H_L + H_R + B + B^\dagger, \\ H_X &= \epsilon_X A_X + \frac{U_X}{2} A_X(A_X - 1), \\ B &= \sum_{i<j} g_{ij} F^\dagger_{ij,L} F_{ij,R}. \end{aligned} \tag{85}$$

The local terms assign an area energy and penalize concentration of area on a copy. This repulsion acts on the whole polyhedron's area, rather than on occupations at individual faces. The exchange term is quartic in elementary oscillators. Its corresponding face labels specify which pairs communicate; they do not impose geometric sewing or identify a shared face.

The Hamiltonian conserves $A_L + A_R$ and commutes with $\boldsymbol{G}_L$ and $\boldsymbol{G}_R$. The current into the left copy is

$$I_L = \frac{i}{\hbar}[H, A_L] = \frac{i}{\hbar}(B^\dagger - B), \qquad I_R = -I_L. \tag{86}$$

Thus $\langle I_L \rangle = 2\,\mathrm{Im}\langle B\rangle/\hbar$: the transfer is controlled by inter-copy singlet-pair coherence. A definite-area product has zero initial mean current, but correlations and transfer can develop during evolution.

A different interaction would create or annihilate one singlet in each copy,

$$H_{\mathrm{pair}} = \sum_{i<j} \left(\lambda_{ij} F^\dagger_{ij,L} F^\dagger_{ij,R} + \overline{\lambda}_{ij} F_{ij,R} F_{ij,L} \right). \tag{87}$$

It preserves separate closure and $A_L - A_R$, while changing combined area. It would require an energy source in a driven realization and differs from both the same-mode oscillator squeeze and the transfer model calculated here.

B. Geometric preparations and finite-area calculation

To choose FL labels, specify face-area proportions a_i and outward unit normals with $\sum_i a_i \mathbf{n}_i = 0$, then choose spinors satisfying

$$z_i^\dagger z_i = 2a_i, \qquad z_i^\dagger \boldsymbol{\sigma} z_i = 2a_i \mathbf{n}_i. \tag{88}$$

The fixed FL label sets the total quantum area, while relative spinor norms specify its mean proportions. Common rotations are gauge choices. Face-dependent phases must be specified relative to the exchange channels; a common spinor phase merely changes the overall phase of a fixed-area FL ket.

We begin with two regular tetrahedra, then change area or shape separately. The initial state is the ordinary product $|K_L, z_L\rangle \otimes |K_R, z_R\rangle$. The right label is not automatically conjugated: this dynamical preparation is distinct from (60). These pure FL inputs also need not be ground states of (85).

The exact finite support is

$$\mathcal{H}_K = \bigoplus_{k=0}^{K} \mathcal{H}^{\mathrm{singlet}}_{4,k} \otimes \mathcal{H}^{\mathrm{singlet}}_{4,K-k}, \qquad K = A_L + A_R. \tag{89}$$

At $K = 4$ its dimension is 1,210, with local dimensions $1, 6, 20, 50, 105$ at areas $0, 1, 2, 3, 4$. Vacuum endpoints and empty faces are retained. There is no occupation truncation inside this declared support.

The three inputs are regular tetrahedra at $(K_L, K_R) = (2, 2)$ and $(3, 1)$, and a regular/deformed pair at $(2, 2)$. Equal-face-area normals use $x = 1/\sqrt{3}$ and bend angles $\varphi = 90°$ for regular and $30°$ for deformed geometry:

$$\begin{aligned} \mathbf{n}_1 &= (\sqrt{1-x^2}, 0, x), \\ \mathbf{n}_2 &= (-\sqrt{1-x^2}, 0, x), \\ \mathbf{n}_3 &= (\sqrt{1-x^2}\cos\varphi, \sqrt{1-x^2}\sin\varphi, -x), \\ \mathbf{n}_4 &= (-\sqrt{1-x^2}\cos\varphi, -\sqrt{1-x^2}\sin\varphi, -x). \end{aligned} \tag{90}$$

We use unit-norm spinors $(\cos(\vartheta/2), e^{i\chi}\sin(\vartheta/2))$ for each normal; uniform rescaling does not change the normalized fixed-area state. At these small areas the labels characterize preparations, not sharply classical tetrahedra.

All six corresponding face-pair channels have the same real coefficient g, without division by six. Setting $U_L = U_R = U$ and $\epsilon_R = 0$, we compare

$$(g, U, \epsilon_L) = (0,0,0),\ (0.1,0,0),\ (0.1,1,0),\ (0.1,0,2). \tag{91}$$

These are no-exchange, resonant, repulsive and detuned controls. With $\hbar = 1$, exact energy-basis propagation samples 101 times over $0 \le t \le 20$ for all twelve input/interaction combinations. Area probabilities, variance, current, pair coherence, return probability, face areas and flux correlations, positive RS volumes and bipartite entropy are followed together.

C. Transfer, fluctuations and equilibrium comparisons

The unequal regular input exhibits substantial coherent transfer: its resonant left mean area ranges between sampled values approximately 1.0083 and 3.0000. With detuning the range is approximately 2.5120–3.2200 over the same interval. These are sampled values, not analytic extrema. In the equal regular input the mean areas remain equal, but resonant left-area variance reaches approximately 2.722 and the copies become entangled. Equal mean areas therefore do not imply absence of exchange. Repulsion reduces the maximum sampled equal-input variance to approximately 1.050. Volume and face correlations evolve alongside these fluctuations, as shown in Figure 12.

Ground states are calculated separately from the prepared trajectories. All three interacting ground spaces

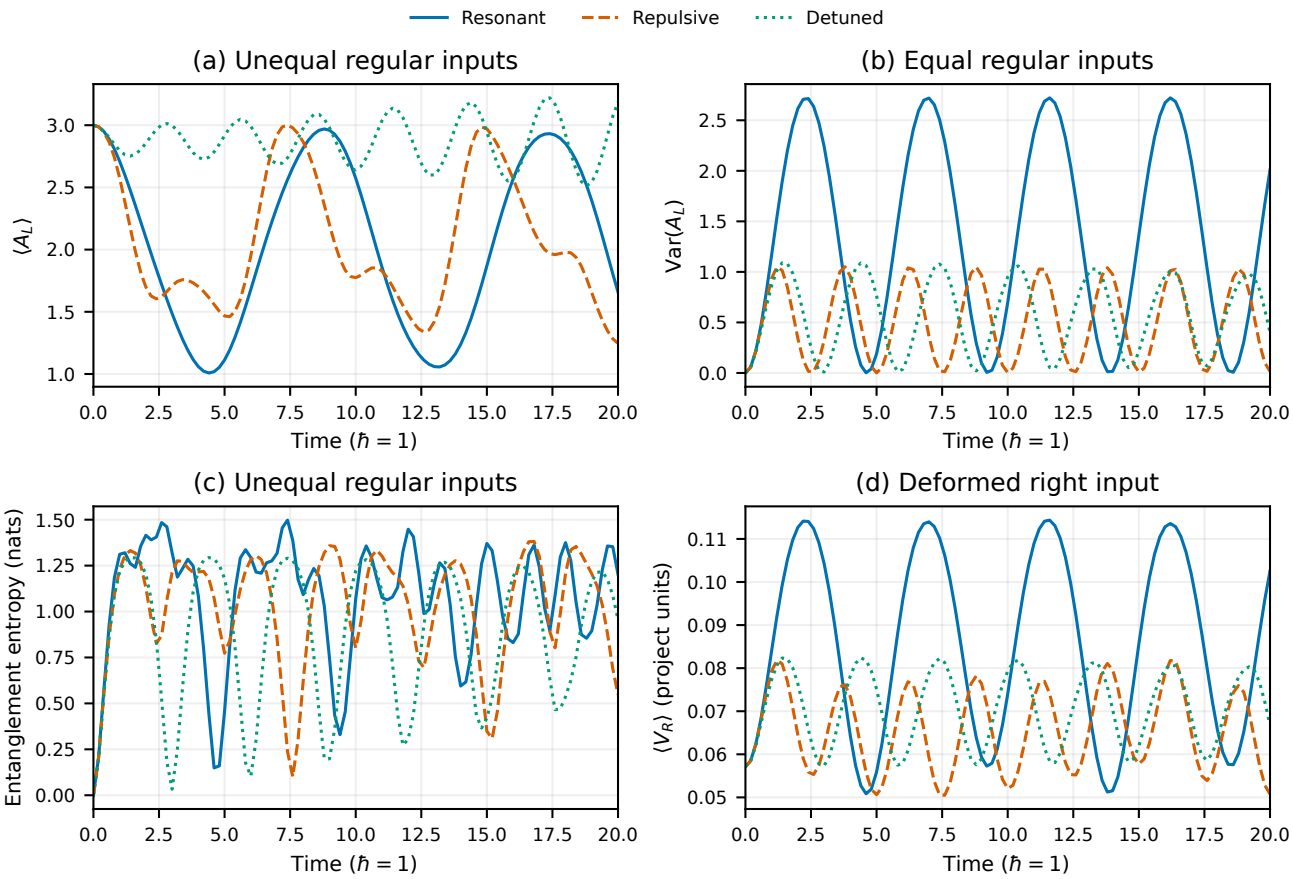


FIG. 12. Singlet exchange between separately closed FL polyhedra at combined area $K = 4$. Panel (a): left mean area for unequal regular inputs. Panel (b): left-area variance for equal regular inputs. Panel (c): entanglement entropy for unequal regular inputs. Panel (d): positive RS volume of the deformed right input. The resonant, repulsive and detuned settings are distinguished by colour and line style. Time uses the specified model energy units with $\hbar = 1$; volume retains repository normalization.

have degeneracy 105, so geometric readouts use their uniform mixtures. Resonant and repulsive ground mixtures have mean left area 2; the detuned ground mixture has mean left area approximately 0.0967. No unique ground-state tetrahedron is inferred. Canonical controls at $\beta = 0, 0.5, 2, 10, 100$ use the same fixed combined-area support. Their joint Gibbs entropy differs from the pure trajectories' bipartite entropy.

An analytic one-area-unit control gives $\langle A_L(t)\rangle = \cos^2(2|g|t)$ for a left-area-one FL input with equal area energies and six equal transfer channels, in units $\hbar = 1$. Direct joint-occupation transfer independently checks the matrix elements and separate-singlet support. Basis closure, creator leakage, Hermiticity, current commutators, norm and energy are checked; the maximum internal residual is 4.53×10^{-14}. Independent sparse propagation of 48 sampled states, together with exact infinite-temperature traces, supplies 1,792 comparisons with maximum residual 9.06×10^{-13} against tolerance 2×10^{-10}.

These results establish coherent exchange and entanglement in a finite isolated pair. The canonical calculations do not show that the FL trajectories thermalize. A reservoir extension must examine recurrence, relaxation and dependence on the initial state before an equilibrium claim can be made.

D. Connections to many-body physics

Singlet transfer has a familiar interpretation in Josephson systems: Cooper pairs tunnel between superconductors through a pair-transfer term, with phase-dependent coherent current. Pair-hopping Bose–Hubbard models provide a bosonic counterpart through $(c_x^\dagger)^2 c_y^2 + \text{h.c.}$ and can support coherence of pairs rather than individual particles [10, 11]. The geometric model replaces that pair by a singlet across two faces, preserving closure during transfer. A Josephson-like area-current interpretation requires physically specified coherence across individual-area sectors, not an arbitrary overall phase of a fixed-area FL ket.

Quantum antiferromagnets supply the closest operator

correspondence. Their Schwinger-boson singlet bond is $a_i b_j - b_i a_j$, the same F_{ij} up to normalization, and antiferromagnetic correlations can be expressed using singlet products [12]. Ordinary spin models fix the occupation at each site; the full intertwiner space permits face occupations and hence areas to vary.

Spinor condensates and optical parametric conversion illustrate the energy source for correlated creation. Spin-changing collisions transfer atoms from an occupied $m = 0$ component into correlated $m = +1, -1$ components [13]. Optical down-conversion uses a pump to produce a signal–idler pair [14]. Treating a pump as classical produces a two-mode squeezing term. Singlet-pair creation would instead excite composite pairs. These correspondences suggest mechanisms to investigate; they do not establish a many-body phase or an equilibrium origin for the projected FL family.

IX. DISCUSSION

The temperature-labelled construction separates ordinary closure from the constraints carried by the transformed and doubled system. Selecting separate singlets retains a closed family with increasing area and positive volume, but the reference eigenvalue ordering excludes a fixed single-copy Gibbs Hamiltonian for that family. Singlet transfer supplies a distinct, closure-preserving mechanism for area dynamics. Its finite-system oscillations, geometric fluctuations and entanglement motivate a reservoir study, rather than establishing thermal relaxation. In a suitable weak-coupling large-reservoir limit, one may compare a copy with $e^{-\beta(H_L-\mu A_L)}/Z$; the area chemical potential reflects combined-area conservation. The reduced spectra and geometric correlations must then be compared with the squeezed-and-projected family, without assuming the two constructions coincide.

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Appendix A: Projected reference coefficients and high-temperature area

Write $K_+ = \sum_i (a_{iL}^\dagger a_{iR}^\dagger + b_{iL}^\dagger b_{iR}^\dagger)$ and $K_- = K_+^\dagger$, so that the common-temperature squeeze is $e^{\theta(K_+ - K_-)}$.

For the initial state $|\phi\rangle_L \otimes |\bar{\phi}\rangle_R$ with $|\phi\rangle = F_{12}^\dagger|0\rangle/\sqrt{2}$, the first two annihilation actions are $K_-|\Phi\rangle = K_{+,12}|0,0\rangle/2$ and $K_-^2|\Phi\rangle = 2|0,0\rangle$, while the third vanishes. Here $K_{+,12}$ restricts the same-mode pair sum to faces 1 and 2. With $\tau = e^{-b/2}$, $\eta = 1-\tau^2$ and final occupation q, combining these coherent contributions yields

$$\begin{aligned} C_q &= \eta^4 \tau^{q-2}(uX + vn_{12} + wI_q), \\ u &= \eta^2/2, \quad v = -\eta\tau^2/2, \quad w = \tau^4. \end{aligned} \tag{A1}$$

Separate singlet projection retains $q = 2K$ and gives (78). The factors in this expression are amplitudes; their interference must be retained before taking squared coefficients.

Couple faces 1 and 2 to spin s and faces 3 and 4 to the same spin, then couple the two to zero. Write their occupations as $2s + 2r$ and $2s + 2l$, with $r, l \geq 0$ and $r + l \leq K$. The common coefficient eigenbasis has

$$\begin{aligned} &2s = K - r - l, \\ &f = r(K - l + 1), \qquad a = K + r - l, \\ &X \longrightarrow f, \qquad n_{12} \longrightarrow a, \end{aligned} \tag{A2}$$

and multiplicity $(K - r - l + 1)^2$. Squaring $uf + va + w$ gives the six terms $f^2, a^2, 1, fa, f, a$. Their multiplicity-weighted sums are finite polynomials in K, which determine retained probability and the unnormalized area moment analytically.

Using

$$\sum_{K=0}^{\infty} K^d e^{-2bK} \sim \frac{d!}{(2b)^{d+1}} \qquad (b \to 0) \tag{A3}$$

for the leading polynomial terms gives $p_\beta \sim 7b^3/256$ and $R_\beta \sim 57b^2/512$. Their ratio yields (73). This asymptote is obtained from exact sector sums rather than the finite occupation display or numerical volume cutoff.

Appendix B: Derivation of Thermofield Double State

The material in this section is standard and is included here to make the article self-contained for most authors. We follow [1, Sec 5.1] in the following. We want to find a thermal state $|0(\beta)\rangle$ which satisfies the following condition: *the expectation value of any operator in* $|0(\beta)\rangle$ *is identical to the statistical ensemble average of the matrix elements of that operator.* This can be written as:

$$\langle A\rangle = \langle 0(\beta)|A|0(\beta)\rangle = \frac{1}{Z}\sum_n e^{-\beta E_n}\langle n|A|n\rangle \tag{B1}$$

To proceed, let us assume that $|0(\beta)\rangle$ is a vector in the original Hilbert space $\mathcal{H}_0$ whose basis is given by the states $\{|n\rangle, n \in \mathbb{Z}\}$, in which case $|0(\beta)\rangle$ can be expanded in this basis:

$$|0(\beta)\rangle = \sum_n c_n |n\rangle$$

Substituting this expression back in (B1) we find:

$$\langle 0(\beta)|A|0(\beta)\rangle = \sum_{m,n} c_m^* c_n \langle m|A|n\rangle = \frac{1}{Z}\sum_n e^{-\beta E_n}\langle n|A|n\rangle$$

The second equality on the right will hold if the coefficients c_n satisfy an orthogonality condition:

$$c_m^* c_n = \frac{1}{Z}\sum_n e^{-\beta E_n}\delta_{mn} \tag{B2}$$

However, such a condition cannot be satisfied if the coefficients c_n are scalars ($\mathbb{C}$ numbers in this case). This implies that the state $|0(\beta)\rangle$ cannot be an element of the original state space $\mathcal{H}_0$. The above condition *would* be satisfied in the coefficients c_n were vectors instead of scalars. Though one could take the representative set of vectors to be any orthogonal basis of $\mathcal{H}_0$, it turns out to be simplest to identify the c_n with elements of the energy eigenbasis:

$$c_n \to |c_n\rangle = d_n(\beta)|n\rangle$$

Inserting this expression into (B2) we see that the state $|0(\beta)\rangle$ can be written as:

$$|0(\beta)\rangle = \sum_n d_n(\beta)|n\rangle|\tilde{n}\rangle$$

This is no longer a state in the Hilbert space $\mathcal{H}_0$ we started with, but in the "double copy" $\mathcal{H} = \mathcal{H}_0 \otimes \mathcal{H}_0$ space. We have put a $\tilde{}$ over the second occurrence of n to indicate that it refers to states in the second copy of the Hilbert space. Since the thermofield double does not live in the same Hilbert space as the original system, the natural question is how can one calculate thermal expectation values for operators which act on the original system using the thermofield state? We can do this as follows. Given any operator $\hat{A}$ which acts on $\mathcal{H}_0$, we can extend this to an operator $\hat{\mathcal{A}}$ which acts on $\mathcal{H}$, via:

$$\hat{A} \to \hat{\mathcal{A}} = \hat{A} \otimes \mathbb{1}. \tag{B3}$$

If we now evaluate the expectation value of $\hat{\mathcal{A}}$ on the state $|0(\beta)\rangle$ defined in (B3) we find:

$$\begin{aligned}\langle 0(\beta)|\hat{\mathcal{A}}|0(\beta)\rangle &= \sum_{m,n} d_m^*(\beta)d_n(\beta)\langle m|\hat{A}|n\rangle\langle\tilde{m}|\tilde{n}\rangle \\ &= \sum_n |d_n(\beta)|^2\langle n|\hat{A}|n\rangle\end{aligned} \tag{B4}$$

Note that, as before, $\tilde{m}$ and $\tilde{n}$ are numerically equal to m and n. Requiring the above expression to match the r.h.s of (B1) implies that the coefficients are given by:

$$d_n(\beta) = \frac{e^{-\beta E_n/2}}{\sqrt{Z}},$$

which brings us to our final expression for the thermofield double state for a harmonic oscillator at an inverse temperature β:

$$|0(\beta)\rangle = \frac{1}{\sqrt{Z}}\sum_n e^{-\beta E_n/2}|n,\tilde{n}\rangle \tag{B5}$$

Appendix C: Unitary Operator for Thermal States

Recalling that the number operator states can be written in terms of the action of ladder operators (12) on the vacuum $|0\rangle$, the thermofield state can be written as:

$$\begin{aligned}|0(\beta)\rangle &= \frac{1}{\sqrt{Z}}\sum_n e^{-\beta E_n/2}\frac{(a_L^\dagger)^n}{\sqrt{n!}}\frac{(a_R^\dagger)^n}{\sqrt{n!}}|0,\tilde{0}\rangle \\ &= \frac{1}{\sqrt{Z}}\sum_n \frac{\left(e^{-\beta\hbar\omega/2}a_L^\dagger a_R^\dagger\right)^n}{n!}|0,\tilde{0}\rangle \\ &= \frac{1}{\sqrt{Z}}\exp\left(e^{-\beta\hbar\omega/2}a_L^\dagger a_R^\dagger\right)|0,\tilde{0}\rangle,\end{aligned} \tag{C1}$$

where in the second line we have used the fact that for a harmonic oscillator the energy eigenvalues are given by $E_n = n\hbar\omega$. In the final step we note that the sum over the coefficients in the second step is the Taylor series expansion of the exponential of an operator $e^{-\beta\hbar\omega/2}a_L^\dagger a_R^\dagger$. In fact one can show (see *e.g.* , [1, Sec 6]) that the thermal state can be obtained from the doubled copy ground state by the action of a unitary operator:

$$|0(\beta)\rangle = e^{-iG(\theta)}|0,\tilde{0}\rangle \tag{C2}$$

where the operator $G(\theta)$ is given by:

$$G(\theta) = -i\theta(\beta)(a_L a_R - a_L^\dagger a_R^\dagger). \tag{C3}$$

Here $\theta(\beta)$ is related to the inverse temperature β by:

$$\theta(\beta) = \tanh^{-1}(e^{-\beta\hbar\omega/2}) \tag{C4}$$

It can be easily seen that $G(\theta)$ is a Hermitian operator and therefore $e^{-iG(\theta)}$ is a unitary operator.

Appendix D: Spin Coherent State Projection

In order to implement the constraint $2j = n_a + n_b$ in (33) we first rewrite that expression as a sum over j, m instead of a sum over n_a, n_b:

$$\begin{aligned}|z_a, z_b\rangle &= e^{-(|z_a|^2+|z_b|^2)/2}\sum_{n_a,n_b}\frac{z_a^{n_a}z_b^{n_b}}{\sqrt{n_a!n_b!}}|n_a,n_b\rangle \\ &= e^{-(|z_a|^2+|z_b|^2)/2}\sum_{j=0}^{\infty}\sum_{m=-j}^{j}\frac{z_a^{j+m}z_b^{j-m}}{\sqrt{(j+m)!(j-m)!}}|j,m\rangle \\ &= e^{-(|z_a|^2+|z_b|^2)/2}\sum_{j=0}^{\infty}z_a^j z_b^j\sum_{m=-j}^{j}\frac{z_a^m z_b^{-m}}{\sqrt{(j+m)!(j-m)!}}|j,m\rangle,\end{aligned} \tag{D1}$$

where in the second line we have simply rewritten the sum over n_a, n_b as being over j, m and in the next step we have pulled out factors of z_a, z_b which depend on j. Now define $z = z_b/z_a$. This implies $z_b = z z_a$ and:

$$z_a^{j+m}z_b^{j-m} = z_a^{j+m}z^{j-m}z_a^{j-m} = z_a^{2j}z^{j-m}$$

Using this the we can write the projected spin coherent state as:

$$P_j|z_a, z_b\rangle = e^{-(|z_a|^2+|z_b|^2)/2} z_a^{2j}|j; z\rangle, \tag{D2}$$

where $|j; z\rangle$ is given by:

$$|j; z\rangle = \sum_{m=-j}^{j} \frac{z^{j-m}}{\sqrt{(j+m)!(j-m)!}}|j; m\rangle. \tag{D3}$$

Appendix E: LQG Phase Space

The Hilbert space consists of cylindrical functions $\psi(h_1, \ldots, h_{N_e})$ where $h_1, \ldots, h_{N_e}$ are $SU(2)$ valued holonomies of the Ashtekar-Barbero connection along each edge. The (kinematic) Hilbert space of a graph thus consists square integrable functions on N_e copies of $SU(2)$:

$$\mathcal{H}_\Gamma = \mathcal{L}^2(SU(2)^{N_e}) \tag{E1}$$

We can use the Peter-Weyl theorem to write functions on $SU(2)$ in terms of spin labels (representations of $SU(2)$) living on each edge:

$$\psi(g) = \sum_{j;mn} f_j^{mn} D^j_{mn}(g) \tag{E2}$$

where $\psi(g) : SU(2) \to \mathbb{C}$ is a function of the holonomy, j labels the different irreps of $SU(2)$, D^j_{mn} are the matrix elements of the group element g in the irrep j and f_j^{mn} are the coefficients of the function in the spin basis. Now each matrix D^j is $(2j+1)\times(2j+1)$ matrix and therefore can be viewed as a map from one copy of $\mathcal{H}_j$ to another copy:

$$D^j : \mathcal{H}_j \to \mathcal{H}_j \tag{E3}$$

Alternatively D^j can also be viewed as an element of $\mathcal{H}\otimes\mathcal{H}$. Therefore the space of square-integrable functions on $SU(2)$ can be written in terms of a direct sum of all irreps of two copies of $SU(2)$:

$$\mathcal{L}^2(SU(2)) = \oplus_j(\mathcal{H}\otimes\mathcal{H}) \tag{E4}$$

This is what leads us to what will be the most crucial point going forwards: *with each edge there are associated two copies of the Hilbert space of a spin j particle.* This is shown in the upper half Figure 7.

Appendix F: Spinor Notation

In this section we present some basic notations and relations involving spinors [15] which are required for understanding some of the expressions in the main text. A spinor $|z\rangle$ is an element of $\mathbb{C}^2$, while $\langle z|$ is its hermitian adjoint:

$$|z\rangle = \begin{pmatrix} z^0 \\ z^1 \end{pmatrix}; \quad \langle z| = \begin{pmatrix} \bar{z}^0 & \bar{z}^1 \end{pmatrix} \tag{F1}$$

The "dual spinor" $|z]$ is defined in terms of the spinor $|z\rangle$ as follows:

$$|z] = \epsilon|\bar{z}] = \begin{pmatrix} -\bar{z}^1 \\ \bar{z}^0 \end{pmatrix} \tag{F2}$$

where ϵ is the antisymmetric tensor with components:

$$\epsilon = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \tag{F3}$$

The combination of complex conjugation $|z\rangle \to K|z\rangle = |\bar{z}\rangle$ followed by multiplication by ϵ can be written as the map: $J = \epsilon K$ which satisfies $J^2 = -\mathbb{1}$. There are two different spinor bilinears:

$$\begin{aligned} \langle x|y\rangle &= [y|x] = \bar{x}^0 y^0 + \bar{x}^1 y^1; \\ [x|y\rangle &= -[y|x\rangle = x^0 y^1 - x^1 y^0 \end{aligned} \tag{F4}$$

From any spinor $|z\rangle$ one can construct a Hermitian matrix which can be expanded in terms of the $(\mathbb{1}, \vec{\sigma})$ basis as follows:

$$|z\rangle\langle z| = \frac{1}{2}(\langle z|z\rangle \mathbb{1} + \vec{X}\cdot\vec{\sigma}) \tag{F5}$$

where $\vec{X} \in \mathbb{R}^3$ and $\|X\| = \langle z|z\rangle$. The above map is not one-to-one because a $U(1)$ rotation $|z\rangle \to e^{i\theta}|z\rangle$ leaves the matrix $|z\rangle\langle z|$, and hence the vector $\vec{X}$, invariant. As mentioned in the text we can construct an element $g \in SU(2)$ from two spinors via:

$$g(z, \tilde{z}) = \frac{|z\rangle[\tilde{z}| - |z]\langle\tilde{z}|}{\|z\|\|\tilde{z}\|} \tag{F6}$$

The inverse of this element is equal to its Hermitian adjoint by unitarity of $SU(2)$ and is given by:

$$g^{-1}(z, \tilde{z}) = g^\dagger(z, \tilde{z}) = \frac{|\tilde{z}]\langle z| - |\tilde{z}\rangle[z|}{\|z\|\|\tilde{z}\|} \tag{F7}$$

It is easy to verify using the spinor bilinear identities given in (F4) that indeed: $g^{-1}g = \mathbb{1}$. It takes a little more work to verify that $\det(g) = 1$. The action of the holonomy on spinors (when the area matching constraint (45) is satisfied) is given by:

$$g\frac{|\tilde{z}\rangle}{\|\tilde{z}\|} = -\frac{|z]}{\|z\|}; \quad g^{-1}\frac{|z\rangle}{\|z\|} = \frac{|\tilde{z}]}{\|\tilde{z}\|} \tag{F8}$$

Using (44) one can verify that under the *local* action of $SU(2)$ on a spinor pair:

$$(|z\rangle, |\tilde{z}\rangle) \to (h_1|z\rangle, h_2|\tilde{z}\rangle), \tag{F9}$$

the holonomy $g(z, \tilde{z})$ transforms exactly as a holonomy should:

$$g(z, \tilde{z}) \to h_1 g(z, \tilde{z}) h_2^{-1}, \tag{F10}$$

where $h_1, h_2 \in SU(2)$ are gauge transformations acting on the source and target vertices of a given edge respectively. One can verify that when (45) holds, the $\mathfrak{su}(2)$ element $X = \vec{X} \cdot \vec{\sigma}$ (corresponding to $|z\rangle$) and the element $\tilde{X} = \vec{\tilde{X}} \cdot \vec{\sigma}$ (corresponding to $|\tilde{z}\rangle$) satisfy:

$$\tilde{X} = -g^{-1} X g \tag{F11}$$

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